

Tutorial FMEA

1. Statement



In order to develop the maintenance process plan of the machining centres in your workshop, in this context you will carry out the FMEA of the centralized lubrication system of the machine slides and will cover the malfunctions of the system during normal operation.

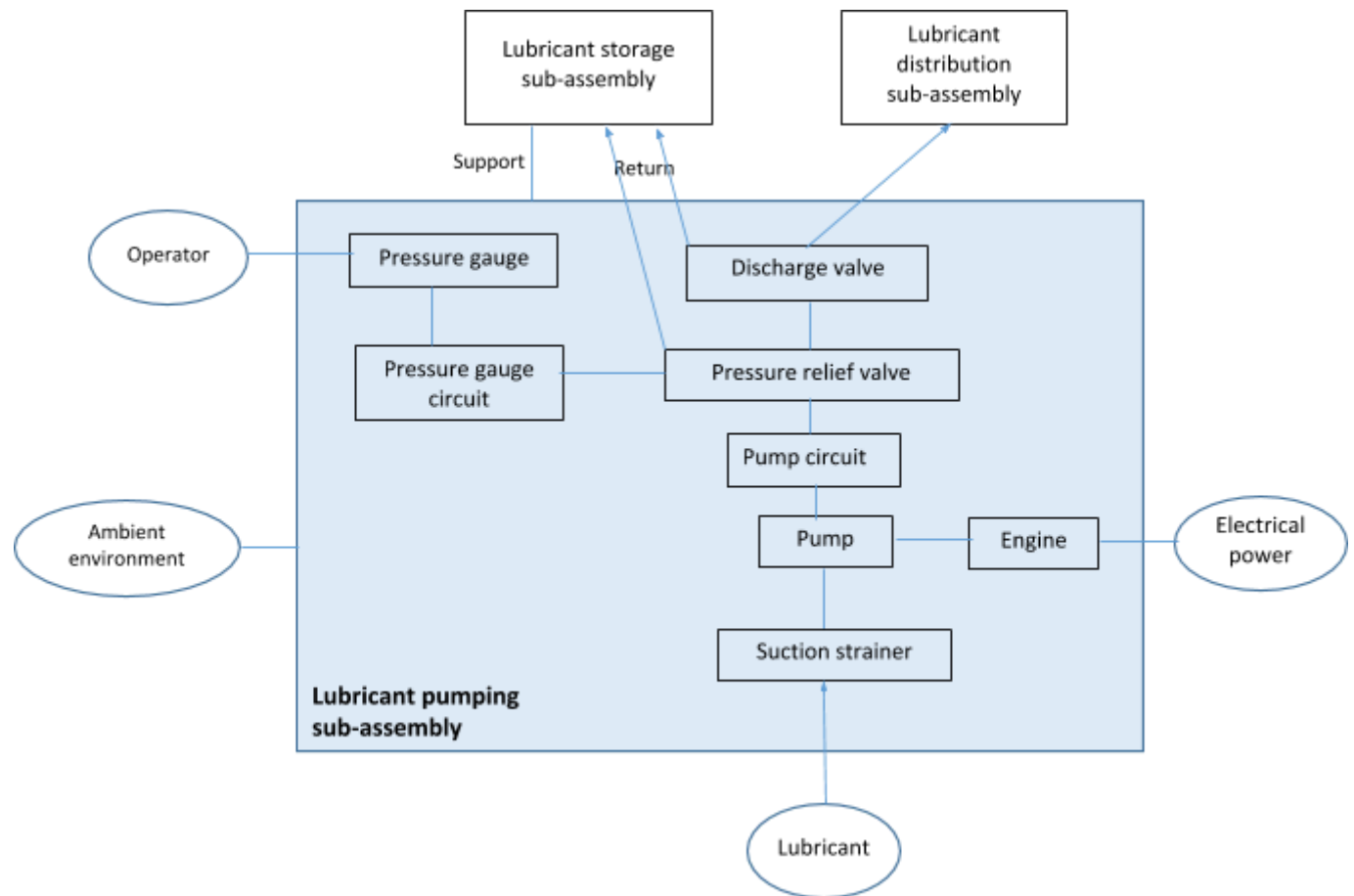
The centralised lubrication system consists of a pumping unit which feeds the distributor-dispensers via a primary circuit. These dispensers deliver the

lubricant to the components to be lubricated via secondary circuits.

The lubrication cycles are triggered by the numerical control of the machine tools. The system reservoir is equipped with a sealed contact level to detect a lack of lubricant.

A pressure switch at the end of the primary circuit allows the detection of insufficient pressure at the end of the lubrication cycle. This pressure switch can be used to stop the machine tools and display an alarm message on the machine tool screen.

Functional Diagram:



2. Goal

Your objective is to carry out a FMEA (or FMECA) of the centralized lubrication system for machine guideways

Reminder:

Summary and Actions to be taken depending on the level of criticality

RPN (O x S x D)	Example of corrective action to be taken
$1 < \text{RPN} < 12$ Negligible Criticality	No design changes, corrective maintenance
$12 \leq \text{RPN} < 16$ Average Criticality	Element Performance Improvement, Systematic Preventive Maintenance
$16 \leq \text{RPN} < 20$ High Criticality	Revision of the design of the sub-assembly and the choice of components, special surveillance, conditional preventive maintenance
$20 \leq \text{RPN} < 80$ Forbidden Criticism	Complete questioning of the design

		Failure Mode and Effects Analysis												FMEA MACHINE			
		System:				Operating phase:											
		Sub-System:				Date of analysis:								Page:			
ELEMENT	FUNCTION	FAILURE			DETECTION	CRITICITY				EVOLUTION							
		MODE	EFFECT	CAUSE		O	S	D	RPN	CORRECTIVE ACTION	RESP.	DEADLINE	nO	nS	nD	New RPN	

Corrective Actions:

QPM: Quarterly Preventive Maintenance

SAPM: Semi-Annual Preventive Maintenance

APM: Annual Preventive Maintenance

MR: Modification to be carried out

SP: Spare part

M: Miscellaneous

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